News Shocks in Prediction Markets

Real results, 2026-10-13

Avi Kabra

Context

- Does company news predictably move Polymarket company-stock contracts? No existing tool answers this directly, so this project builds two new methods.
- Data: 6,928 resolved contracts across 86 companies (Apr 2024-Aug 2026), matched against 546,657 GDELT articles
- GDELT only timestamps articles to the day, so every result below measures the 24-hour price reaction, the only window this feed supports

Method

Matching picks the right article for a contract, combining text similarity and trading-volume spikes:

match_quality = w1 · logit(cosine similarity) + w2 · z(volume spike), then sigmoid

Consistency forces a company's related contracts (above $150, $160, $170) to react in the same direction, weighted by volume, via isotonic regression (PAVA).

Both are new methods built for this problem, not off-the-shelf tools, and both are validated on real held-out data below.

Result 1: the matching method works

Method	Spearman ρ vs. actual price move	n
Matching score (text + volume)	**0.509**	14,717
Text similarity alone	0.233	14,717

Given a list of candidate articles for a contract, this score reliably tells us which one actually explains the price move. Text alone is close to a coin flip; the combined score is not.

Result 2: the consistency correction doesn't help, so we don't use it

Category	R^2, raw reaction	R^2, after correction
Price-target	0.084	0.073
Valuation	0.746	0.363

The correction shifts values by a real amount but makes predictions worse in both testable categories. We predict from the raw reaction, not the corrected one.

Result 3: none of the four models give a usable edge

Category	Best model	R^2	n (test)
All (pooled)	Linear	−0.010	5,151
Corporate event	Linear	−0.021	3,157
Price-target	LSTM	**+0.010**	598
Revenue-target	TCN	−0.016	9
Valuation	Linear only	−0.255	small

No category clears a plain guess-the-average baseline by a meaningful margin. Revenue-target is trained here for the first time, on its own data split, and still lands negative.

Result 4: one of four pre-registered predictions holds up

Prediction	Result
Richer embeddings beat simpler ones	Not supported
Best model differs by category	Not supported
Quantitative articles beat high-attention ones	Supported (n=28)
Best ladder category beats corporate events	Not supported

Tested with bootstrap resampling clustered by real-world event, so duplicate coverage of one story doesn't count as independent evidence. Only the third prediction survives, on a small sample.

Scoping decisions

- **market_cap_ladder** (4 markets): dropped, too small to mean anything
- **valuation_ladder** (110 markets): kept, flagged as below our 200-market floor, everything else about it clears comfortably
- **revenue_ladder**: given its own data split so it could be trained at all, rather than excluded

Limitations

- GDELT is free but noisy, and day-precision is a structural limit of this feed, not a gap to close
- `resolved_at` and `parent_event_id` are each about 40% backfilled from Polymarket's API; fallbacks cover the rest
- The revenue-target result rests on 19 training examples: real, but small

Recommendation

- Lead the thesis with the matching result (Result 1): it is the clean, strong, genuine contribution
- Present the consistency correction and the model comparison as honest negative results, not failures
- Keep the 200-market floor as the rule, with `valuation_ladder` as a documented, explicit exception
- A licensed newswire is the single highest-leverage next step if budget allows; everything else here is close to final